MID-TERM PROJECT UPDATE

Transformer for Time Series Anomaly Detection

Building and evaluating a compact, efficient model for temporal data

GROUP 7

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OUR GOAL

To develop a transformer-based model benchmarked against strong classical & neural baselines

Problem & Current Focus

- **Task:** Univariate time series anomaly detection
- **Dataset:** NAB NYC Taxi (trips every 30 mins)
- **Characteristics:** Strong daily/weekly seasonality with rare anomalies (e.g., holidays, marathon)
- **Why this setup?**
  A well-known benchmark simplifies comparison and serves as an ideal sandbox for testing evaluation pipelines

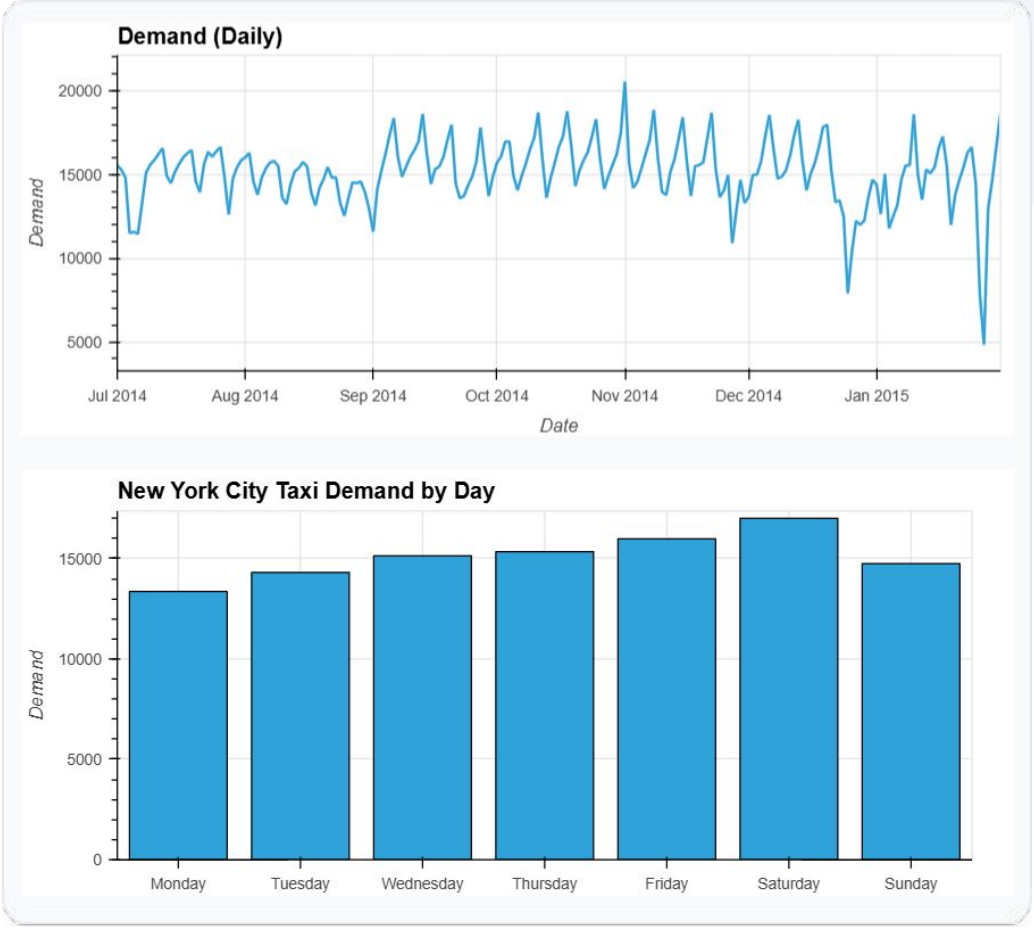


Figure: Weekly seasonality in NYC taxi demand

Data & Experiment Pipeline

Data Handling

- Implemented a data loading and preprocessing pipeline
- Time-based split (70/30) with correct DatetimeIndex parsing ensures no data leakage

Experiment Automation

- Automated train/test/scoring pipeline for each model
- Ground-truth labels integrated for evaluation
- Designed for Dockerization and reproducible experiments

Evaluation Metrics

- Using Precision, Recall, F1, and AUC-ROC
- Added Range Metrics to handle contiguous anomaly intervals more fairly

Classical Baselines Implemented

Rolling Z-Score

Uses a 1 day window. Large Z-scores are flagged. While simple, it is highly sensitive to threshold settings and often weak

ARIMA Residuals

ARIMA(2,1,2) one-step forecast. Anomalies are large residuals. The scores are informative but noisy, making thresholding tricky

Isolation Forest

Tuned with contamination ≈ 0.01. It detects 2/3 of labeled events with the best F1 range and AUC-ROC. **Current strongest classical baseline**

Neural Baseline & Transformer Work

LSTM Forecasting Baseline

- **Architecture:** Univariate LSTM with a 72-step lookback window predicting the next value
- **Scoring:** Anomaly score = Absolute forecast error
- **Behavior:** Very few alarms → high precision but lower recall compared to Isolation Forest

Transformer Progress

- Mini-transformer built from scratch (encoder-style)
- **TranAD** deployed and running on University scientific cluster for heavy runs

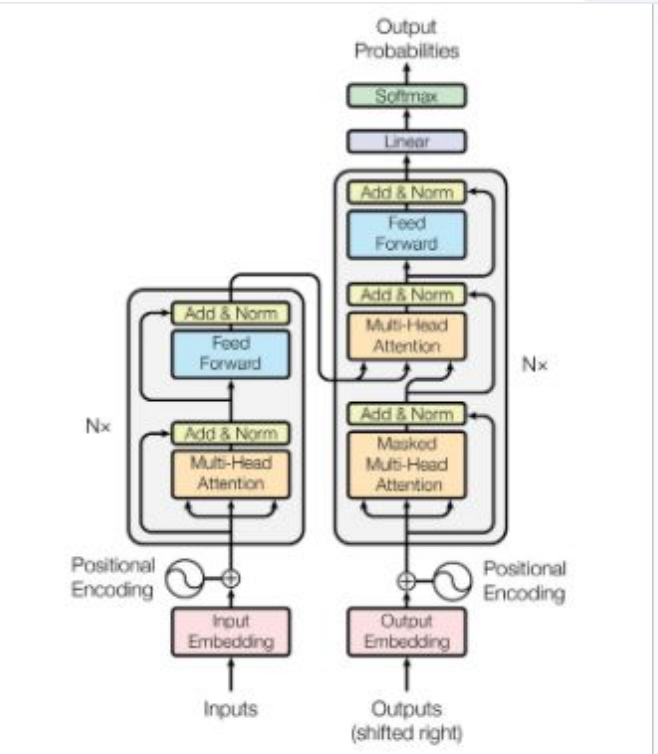


Figure 1: The Transformer - model architecture.

Figure: Transformer Architecture (Reference)

Key Insights So Far

Metrics & Labels

- Given the few labeled anomalies, point metrics are unstable
- **Range-based metrics** provide a more intuitive and fair evaluation of model performance

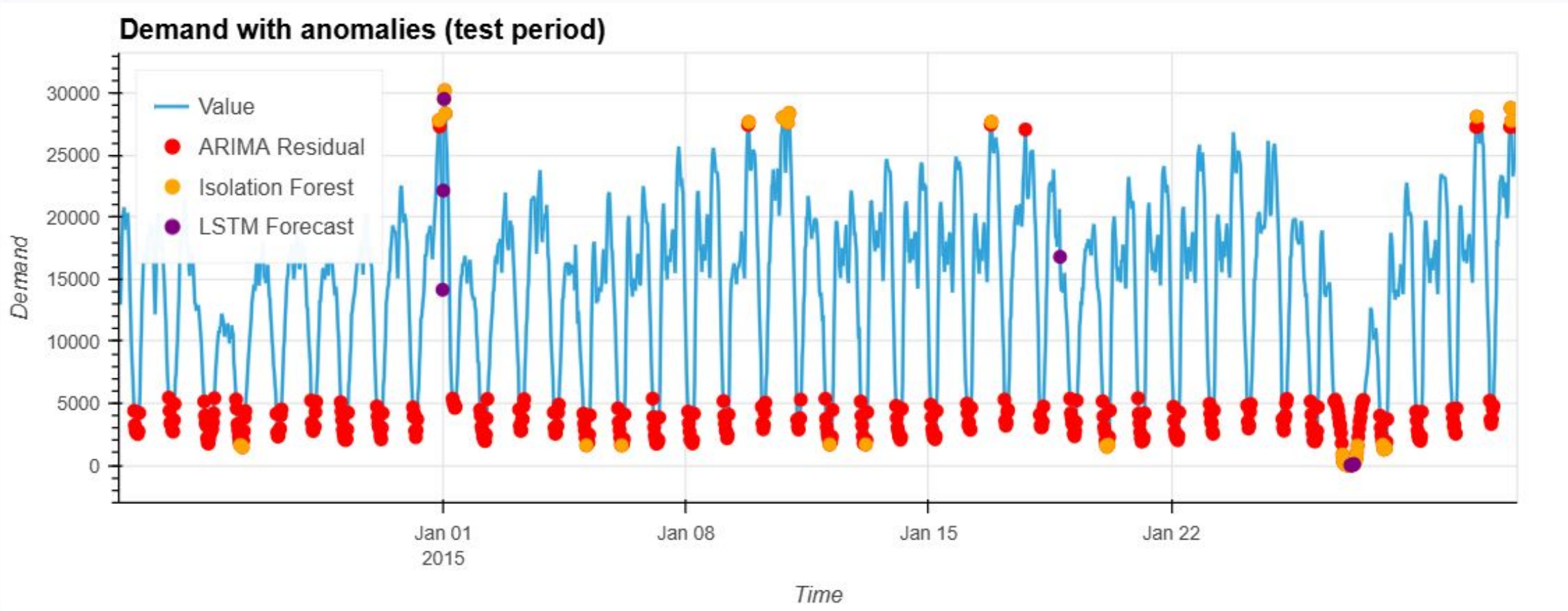
Threshold Sensitivity

- Small changes in thresholds (Z-score limits, contamination rates) lead to massive swings in precision/recall
- Robust thresholding is a key challenge

Model Roles

- **ARIMA** is too noisy for this dataset
- **LSTM** is conservative (high confidence, low quantity)
- **Isolation Forest** offers the best trade-off so far

Baseline Anomaly Detection



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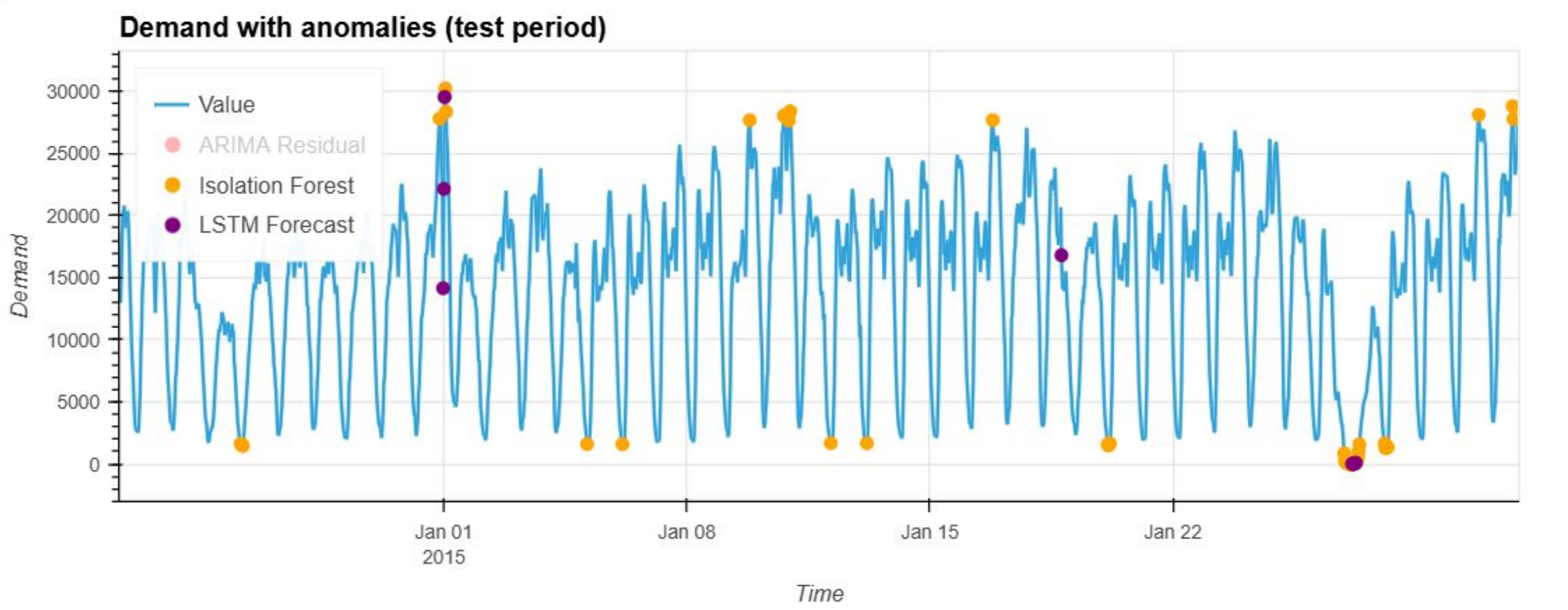

Next Steps

Transformer Experiments

- Train and tune transformer on NYC Taxi & additional NAB datasets
- Find the right embeddings: **Sinusoidal vs. Time2Vec**
- Test strategies: Reconstruction vs. Forecasting vs. Hybrid scoring

Stronger Baselines & Outcome

- Add features (lag, hour-of-day, day-of-week) to Isolation Forest & LSTM
- Fair comparison of classical, LSTM and transformer-based methods
- **Goal:** End-to-end demo with visual anomaly overlays and a fair performance summary